

Garrett Van Dyke

🔒 TS/SCI, Polygraph | ✉️ garrett.vandyke@icloud.com | 📞 904-486-8265

SKILLS

Languages: Python (*dask, mpi4py, pydantic, polars, FastAPI, matplotlib, networkx, numpy, pandas, PyQT, pyzmq, scipy*), MATLAB, Java, TypeScript, LaTeX, Mathematica

Platforms and Operating Systems: GitLab, Jira, Confluence, GitHub, Red Hat Enterprise Linux 8, Windows 10 & 11

EXPERIENCE

Northrop Grumman

2024 – Present

Modeling & Simulation Software Development Lead, Principal Physicist

Linthicum, MD

- Designed, co-authored, and maintained package for running parallel (MPI) quantum circuit simulation workflows on an HPC system. Led scrum meetings, contributed high-level designs, reviewed code, and organized priorities with stakeholders.
- Managed development standards for group of 50+ contributors/users. Enforced standards through GitLab CI/CD pipeline configurations with varying strictness based on team need.
- Co-authored and maintained launcher for running compressed python environments on an HPC system.

Northrop Grumman

2022 – 2024

Test Capabilities Software Developer, Physicist

Linthicum, MD

- Co-authored GUI (PyQT) for cryogenic data acquisition framework. Included a workflow viewer, a database viewer, an instrument control interface, interactive pulse timing diagrams, and a real-time plotter.
- Wrote GUI (bokeh) enabling users to query chip fabrication performance metrics.
- Wrote MATLAB-to-Python interface allowing testers to hook custom Python analyses into a MATLAB data acquisition suite.

Quanterix Corp.

Summer 2021

Data Science Intern

Billerica, MA

- Maintained Java codebase for the SP-X planar immunoassay imaging and analysis platform. Provided bugfixes, image analysis improvements, and an automated regression test report generator for codebase validation.

University of Alabama at Birmingham

2019 – 2021

Graduate Researcher

Birmingham, AL

- Assisted Dr. Ryoichi Kawai in computational physics research focusing on fundamental thermodynamics of open quantum systems. Used the Hierarchical Equations of Motion method on HPC system to simulate time evolution of a density matrix in a lossy environment.

NASA Marshall Space Flight Center

Summers 2018 & 2019

Systems Engineering Intern

Huntsville, AL

- Developed tools to analyze test data from the RS-25 rocket engine in Python and VBA.

EDUCATION

The University of Alabama at Birmingham

Aug. 2019 – Dec. 2021

Master's of Science in Physics, Open Quantum Systems

Birmingham, AL

- Blazer Research Fellow

Samford University

Aug. 2015 – May 2019

Bachelor of Science in Physics, Mathematics Minor

Birmingham, AL

- University Fellow